4 (a)

for log. Regression likelihood function

$$L(w) = \sum_{n=1}^{N} \left\{ y_i \log \left(\frac{1}{1+\exp(-w^T x)} \right) + (1-y_i) \log \left(1 - \frac{1}{1+\exp(-w^T x)} \right) \right\}$$

(ii) gradient

$$\nabla_w L(w) = \sum_{n=1}^{N} \left\{ 1 - \frac{y_i}{1+\exp(-w^T x)} \right\} . x$$

(iii) hessian

$$\nabla_w^2 L(w) = \sum_{n=1}^{N} \left[\frac{-\exp(-w^T x)}{(1+\exp(-w^T x))^2} \right] . x_i . x_i^T$$

(iv) update equation

$$W_{new} = W_{old} - \frac{\nabla_w L(w)}{\nabla_w^2 L(w)}$$

$$W_{new} = W_{old} - \frac{\sum_{n=1}^{N} \left\{ 1 - \frac{y_i}{1+\exp(-w^T x)} \right\} x}{\sum_{n=1}^{N} \left\{ \frac{-\exp(-w^T x)}{(1+\exp(-w^T x))^2} \right\} x_i . x_i^T}$$

Algorithm of Newton-Raphson Optimization technique:-

(A) Initialize the parameter vector W

Convergence threshold, $\epsilon = 0.001$

no. of Iterations T_{max}

change in Weights $\Delta W = \infty$

current iteration number, $t = 0$

(B) while $t < T_{max}$ && $\|W\| > \epsilon$:

(i) Compute $\underset{w}{\Delta} L(w) = \sum_{n=1}^{N} \left(y_i - \frac{1}{1+\exp(-w^T x)} \right)$

$$\underset{w}{\Delta}^2 L(w) = \sum_{n=1}^{N} \left(\frac{-\exp(-w^T x)}{(1+\exp(-w^T x))^2} \right) \cdot x_i \cdot x_i^T$$

(ii) update weight by

$$W_{new} = W_{old} - \frac{\underset{w}{\nabla} L(w)}{\underset{w}{\nabla}^2 L(w)}$$

(iii) $\Delta W = W_{new} - W_{old}$

(iv) $W_{old} = W_{new}$

(v) $t = t + 1$

(C) Return W_{new}